Transposed Hexanies: Just intonation synthesis studies

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Just intonation tuning systems that have no tonal centres provide the starting point for this discussion on tuning associated with a series of works entitled *Transposed Hexanies*. The composer describes the origins of the hexany, the hexanies used in this these works, their transformation and extension through pitch multiplication and the connections between rhythms and tunings used in these works. The lecture will demonstrate how transposition extends the scope of the hexany. The transposition algorithm will be explained together with its ramifications for live performance using hand-held instruments capable of performing in tunings other than twelve equal divisions of the octave.

1 Motivation : understanding Wilson's Combination Product Sets

Over the past few years, I have been fascinated by the theory of Combination Product Sets, first proposed by Erv Wilson in his article "D'alessandro, Like a Hurricane" in Xenharmonikôn 12 (Wilson 1989) and which he gave me in 1998. I am particularly interested in the unique complementary relationship between major and minor tonality inherent in his theory. I have explored this relationship in a series of eleven studies in just intonation called *Transposed Hexanies*, which I began in May 2000. The first of these studies was premiered at ACMA 2000, Queensland University of Technology, Kelvin Grove on Jul 5th 2000; a further ten studies *Transposed Hexanies* No.2 –No.11 were premiered at MicroFest 2001, Pomona College, Claremont, California on 6th April 2001.

1.1 Combination Product Sets — the hidden tonic

Combination Product Sets are just intonation scales formed when combinations of a set of harmonics are multiplied together. One of the simplest kind is the hexany, a 6-note scale generated by multiplying all combinations of pairs of harmonics from a set of four harmonics. For example, by combining pairs of products the first four odd harmonics 1, 3, 5 and 7, as shown in Table 1a, the pitch products shown in Table 1b are produced. In just intonation, these are expressed as fractions with each pitch (numerator) relative to a tonic (denominator) as shown in Table 1c. Note that the tonic is only implied and is not present in the scale. Pitches have been arranged in ascending order.

My initial observations of the hexany were focused on just major and just minor triads found in the above hexany. In Table 1c, pitches in the odd numbered columns form the just major triad, while pitches in the even numbered columns form the just minor triad.

These complementary major and minor triads are found on diametrically opposed sides of Wilson's geometric representation of the hexany as shown in Figure 1. Wilson's hexany is represented as two super-imposed triangles, one representing tetrads formed by the products of each harmonic, the other its complement formed by the products of each sub-harmonic. Unlike Partch's diamond, which also organises chords around the harmonic and sub-harmonic series with the tonic explicitly included, Wilson's combination product set is "a structure which could produce tonal progression without a tonal centre" (Rapoport 1994). The novelty of the combination product set structure is that the tonic is absent from the scale altogether as represented by the hole at the geometric centre, yet at the same time, it is tonal because of the complementary harmonic relationships it creates.

I wanted to know what sense my ears would make of complementary harmonic relationships in various other hexanies generated using a different set of harmonics. I made a 1 3 5 7 hexany using *Scala*, tuning analysis/editor/librarian software written by Manuel Op de Coul, and Csound, computer music synthesis software written by Barry Vercoe and others. This hexany is heard in the first audio example; the odd numbered notes of the hexany -- ie. a 1st inversion major triad -- are played first, followed by the even numbered notes – a root position minor triad (audio example 1). I played with the Csound code to find out how these major and minor triads sounded in relation to other triadic combinations found within the hexany (audio examples 2a and 2b). I tried hexanies generated using other primes besides 3 5 and 7 but went no further than the prime 11. I restricted myself to hexanies produced from combinations of harmonic generators shown in Table 2. Pitches of the five hexanies are shown in Table 3 (audio example 3).

1.2 Transposition algorithm : the tonic reappears

I also wanted to know what happens if pitches of the hexany are transposed using the intervals of the same hexany. I modified the Csound source code to allow transposition. The Csound Orchestra is shown in Example 1a, the Score in Example 1b.

Transposition introduces interesting possibilities for voice leading, as demonstrated in the following passage from the first study written in May 2000 (audio example 4).

I wrote a scala command file to find out what actual pitches are produced by transposing a hexany. I discovered that transposition replicates some pitches of the original hexany and produces unique pitches ie. pitches that are not found in the original hexany or pitches that are not duplicated in other transpositions. Each hexany when transposed only by itself, produces 19 unique pitches. The total number of unique pitches in the five 6-note scales is 9. The total produced through transposition is 69, a ratio of more than 7 : 1.

Transposing the hexany does more than create additional pitches. By virtue of the hexany's limited set of harmonic relationships, transposition organises a large collection of pitches into smaller manageable groups of pitches. All pitches available through transposition in Hexany 1 are shown in Table 4a; Hexany 2 (4b); Hexany 3 (4c); Hexany 4 (4d) and Hexany 5 (4e). One intriguing thing about these transpositions is that the tonic, or 1/1, which is hidden in the original hexany, re-emerges an octave higher in the first four of these tables. In the following examples all pitches for each transposed hexany are played as complementary major and minor triads (audio examples 5a).

Transposition is implemented in Csound using pitch multiplication. Two linear factors are multiplied, each factor selected from two parameter fields in the Csound score. One parameter field, p5, selects a pitch from the original hexany; the other, p6, selects a transposition interval. In this way, I was able to work with the harmonic features of a single hexany and, quite separately, experiment with its various transpositions.

The Csound implementation of transposition made it a simple matter to transpose the pitches of one hexany, p7, using the intervals of another hexany, p8. This technique, which I will refer to as cross transposition, became the basis of the next ten studies in the collection, *Transposed Hexanies No.2–No.11*.

2 Transposed Hexanies No.2-No.11 (2001)

The ten studies *Transposed Hexanies No.2–No.11* employ all cross-transpositions of five hexanies. Each study is based on one of five hexanies shown in tables 1 and 2. In each study, pitches of one hexany are transposed by intervals of another. A different cross-transposition is used in each study. The ten possible combinations are shown in Table 5.

In each of the ten studies, transposition does not occur continuously throughout the entire study. At some point, each hexany is heard untransposed; elsewhere in the same study, it is transposed at my discretion. This is done in order to modify the harmonic context at various places in each study where I thought some unexpected musical change was warranted. The effect of transposing one hexany with another is analogous to the leaching technique used by artists who work with water colours; new hues of colour are created when a second layer of colour is added while the first layer is still wet.

2.1 Rhythmic templates

In each of the ten studies, a number of rhythmic templates were created using standard music notation software, Finale. I created rhythmic templates using ratios I found in the hexany. Each template is organised in four layers, each layer assigned to a dedicated stave. Each stave corresponds to one of the rhythmic generators used in each study. Each template is saved as a MIDI file which is converted into a Csound score file using MIDI2CS, public domain software developed by Rüdiger Borrmann. The Finale score was never intended as a challenge to a virtuoso performer. It is used solely as a visual means to organise six pitches played as complex rhythms, and enter it into Csound. Once converted to Csound score, I arbitrarily selected the tessitura, transposition and hexany numbers by altering values in parameter fields p5, p6, p7 and p8 of the Csound score. Example 2 shows Hexany 5 in standard notation.

2.2 Gray code sequences

In each of the studies, Gray code is used to create an orderly sequence of states through every combination of rhythmic generator. Gray code creates a transition from one rhythmic template to another whereby only one layer enters or exits at any one time, as shown in Tables 6a-e. Gray code combines these strands of rhythmic information in a single seamless transition from one state to another without any combination being repeated. Each of the four rhythmic templates is heard alone once; each template is heard once in combination with one other template; each template is heard once in combination with two other templates; all templates are heard once together (musical example 6a).

3 Conclusion

Working with the hexany invites a further development of the boolean harmonic filter algorithm I created for the MIDI Tool Box (Schiemer 2000). This algorithm was previously applied in the context of 12-tone equal temperament. It was based on the premise that MIDI note numbers are really two interleaved sets of whole tone scales, one containing odd numbered notes, the other containing even numbered notes. In a just intonation scale like the hexany, where odd numbered notes define major tonality and even numbered notes define minor tonality, the boolean filter algorithm offers a computationally efficient way of manipulating harmonic fields in just intonation.

In its ideal state *Transposed Hexanies* will ultimately be a live perfomance work. At the very least, these studies, implemented in Csound, will provide the basis for a new live performance work. The current realisation is in mono. Some spatialisation will be added in a future implementation. The timbre is currently limited to Csound's pluck instrument. Future enhancements will involve plucked string portamento to give the quality of phrasing created by players of the Vietnamese monochord, known as *dan bau* or the quality of *gamaka* found in Carnatic vocal music.

Ideally, I look forward to a performance of the work as part of a new development project involving battery-operated hand-held instruments. The project will involve performance using current wireless technology based on Java MicroEdition and is based around the Motorola MCore microcontroller. This technology opens possibilities for performing music in ways that are no longer constrained by the current desktop computing paradigm.

Above all, the economy of the transposition technique developed in Transposed Hexanies makes it appropriate in mobile hand-held computing environment. Because it is implemented in a way that combines table look-up with arithmetic performed on-the-fly, it can easily be applied within the hardware

constraints of J2ME technology. Squeezing more out of less is not just a mindless musical pursuit. It is also a way of squeezing more data into small memory address spaces associated with K-Virtual Machine environments. Ultimately the technique could be useful in developing a new generation of mobile hand-held musical instrument capable of performing music tuned in just intonation.

References

Wilson, Ervin 1989 “D’Alessandro, Like a Hurricane” Xenharmonikôn 12 pp. 1-38

Rapoport, Paul 1994 “Just Shape Nothing Centra” Musicworks 60 pp. 42-49

Burt, Warren 1996 "An Interesting Group of Combination-Product Sets Produces Some Very Nice Dissonances" *1/1* 9 [1] p. 1

Schiemer, Greg 2000 “Boolean logic as a harmonic filter” Interfaces Proceedings of Australian Computer Music Conference 2000 pp. 59-67.

(1a)	**5 x 7**	**1 x 5**	**3 x 7**	**1 x 3**	**1 x 7**	**3 x 5**
(1b)	**35**	**5**	**21**	**3**	**7**	**15**
(1c)	**35/32**	**5/4**	**21/16**	**3/2**	**7/4**	**15/8**

Table 1 Derivation of pitches in a hexany

Hexany	Generators			
No. 1	**1**	**3**	**5**	**7**
No. 2	**1**	**3**	**5**	**11**
No. 3	**1**	**3**	**7**	**11**
No. 4	**1**	**5**	**7**	**11**
No. 5	**3**	**5**	**7**	**11**

Table 2 Generators used in the five hexanies

HEXANY 1	**Generators: 1 3 5 7**				**(audio example 3a)**
1: 35/32 septimal neutral second			**5**	**7**
2: 5/4 major third	**1**		**5**	
3: 21/16 narrow fourth		**3**		**7**
4: 3/2 perfect fifth	**1**	**3**		
5: 7/4 harmonic seventh	**1**			**7**
6: 15/8 classic major seventh		**3**	**5**	

HEXANY 2	**Generators: 1 3 5 11**				**(audio example 3b)**
1: 33/32 undecimal comma		**3**		**11**
2: 5/4 major third	**1**		**5**	
3: 11/8 undecimal semi-augmented fourth	**1**			**11**
4: 3/2 perfect fifth	**1**	**3**		
5: 55/32			**5**	**11**
6: 15/8 classic major seventh		**3**	**5**	

HEXANY 3	**Generators: 1 3 7 11**				**(audio example 3c)**
1: 33/32 undecimal comma		**3**		**11**
2: 77/64			**7**	**11**
3: 21/16 narrow fourth		**3**	**7**	
4: 11/8 undecimal semi-augmented fourth	**1**			**11**
5: 3/2 perfect fifth	**1**	**3**		
6: 7/4 harmonic seventh	**1**		**7**	

HEXANY 4	**Generators: 1 5 7 11**				**(audio example 3d)**
1: 35/32 septimal neutral second		**5**	**7**	
2: 77/64			**7**	**11**
3: 5/4 major third	**1**	**5**		
4: 11/8 undecimal semi-augmented fourth	**1**			**11**
5: 55/32		**5**		**11**
6: 7/4 harmonic seventh	**1**		**7**	

HEXANY 5	**Generators: 3 5 7 11**				**(audio example 3e)**
1: 33/32 undecimal comma	**3**			**11**
2: 35/32 septimal neutral second		**5**	**7**	
3: 77/64			**7**	**11**
4: 21/16 narrow fourth	**3**		**7**	
5: 55/32		**5**		**11**
6: 15/8 classic major seventh	**3**	**5**		

Table 3 Pitches of the five hexanies

```
              instr 1

              ihexany    =   p5           ;   select hexany pitch value
              itranspose =   p6           ; select hexany transposition
              ifrq1  cps2pch ihexany,    -(p7)
              ifrq2  cps2pch itranspose, -(p8)
              ifrq       =   ifrq1 * ifrq2
asig          pluck  ampdb(p4), ifrq, ifrq, 0, 1
              aright     =   asig
              aleft      =   asig
              outs aright*(givol*1.00), aleft*(givol*1.00)
              endin
```

Example 1a

```
; ------------------------ Transposed.Hexanies.SCO -------------------

f1 0 8 -2  1.00000000  1.09375000  1.25000000  1.31250000  1.50000000  1.75000000
1.87500000
f2 0 8 -2  1.00000000  1.03125000  1.25000000  1.37500000  1.50000000  1.71875000
1.87500000
f3 0 8 -2  1.00000000  1.03125000  1.20312500  1.31250000  1.37500000  1.50000000
1.75000000
f4 0 8 -2  1.00000000  1.09375000  1.20312500  1.25000000  1.37500000  1.71875000
1.75000000
f5 0 8 -2  1.00000000  1.03125000  1.09375000  1.20312500  1.31250000  1.71875000
1.87500000
; tempo
t 0 15.0000


; p1 = instr
; p2 = start
; p3 = duration
; p4 = dB
; p5 = pitch
; p6 = transposition (nb. octave must = 0)
; p7 = hexany N (pitch)
; p8 = hexany M (transposition)


;p1    p2       p3      p4      p5      p6      p7     p8

i01    0.00000  1.1250  80      8.01    0.01    1      1    ;  D3 ch01
i01    0.00000  1.1250  80      8.03    .       .      .    ;  F3 ch01
i01    0.00000  1.1250  80      8.05    .       .      .    ;  A3 ch01

i01    1.12500  1.0000  79      8.01    .       .      .    ;  D3 ch01
i01    1.12500  1.0000  79      8.03    .       .      .    ;  F3 ch01
i01    1.12500  1.0000  79      8.05    .       .      .    ;  A3 ch01

i01    2.12500  1.0000  78      8.02    .       .      .    ;  E3 ch01
i01    2.12500  1.0000  78      8.04    .       .      .    ;  G3 ch01
i01    2.12500  1.0000  78      8.06    .       .      .    ;  B3 ch01
etc.
```

Example 1b

	Original	Trans. 1	Trans. 2	Trans. 3	Trans. 4	Trans. 5
1:	**35/32**	**5/4**	**21/16**	**3/2**	**7/4**	**15/8**
2:	**5/4**	**10/7**	**3/2**	**12/7**	**2/1**	**15/7**
3:	**21/16**	**3/2**	**63/40**	**9/5**	**21/10**	**9/4**
4:	**3/2**	**12/7**	**9/5**	**72/35**	**12/5**	**18/7**
5:	**7/4**	**2/1**	**21/10**	**12/5**	**14/5**	**3/1**
6:	**15/8**	**15/7**	**9/4**	**18/7**	**3/1**	**45/14**

Table 4a Hexany 1

	Original	Trans. 1	Trans. 2	Trans. 3	Trans. 4	Trans. 5
1:	**33/32**	**5/4**	**11/8**	**3/2**	**55/32**	**15/8**
2:	**5/4**	**50/33**	**5/3**	**20/11**	**25/12**	**25/11**
3:	**11/8**	**5/3**	**11/6**	**2/1**	**55/24**	**5/2**
4:	**3/2**	**20/11**	**2/1**	**24/11**	**5/2**	**30/11**
5:	**55/32**	**25/12**	**55/24**	**5/2**	**275/96**	**25/8**
6:	**15/8**	**25/11**	**5/2**	**30/11**	**25/8**	**75/22**

Table 4b Hexany 2

	Original	Trans. 1	Trans. 2	Trans. 3	Trans. 4	Trans. 5
1:	**33/32**	**77/64**	**21/16**	**11/8**	**3/2**	**7/4**
2:	**77/64**	**539/384**	**49/32**	**77/48**	**7/4**	**49/24**
3:	**21/16**	**49/32**	**147/88**	**7/4**	**21/11**	**49/22**
4:	**11/8**	**77/48**	**7/4**	**11/6**	**2/1**	**7/3**
5:	**3/2**	**7/4**	**21/11**	**2/1**	**24/11**	**28/11**
6:	**7/4**	**49/24**	**49/22**	**7/3**	**28/11**	**98/33**

Table 4c Hexany 3

	Original	Trans. 1	Trans. 2	Trans. 3	Trans. 4	Trans. 5
1:	**35/32**	**77/64**	**5/4**	**11/8**	**55/32**	**7/4**
2:	**77/64**	**847/640**	**11/8**	**121/80**	**121/64**	**77/40**
3:	**5/4**	**11/8**	**10/7**	**11/7**	**55/28**	**2/1**
4:	**11/8**	**121/80**	**11/7**	**121/70**	**121/56**	**11/5**
5:	**55/32**	**121/64**	**55/28**	**121/56**	**605/224**	**11/4**
6:	**7/4**	**77/40**	**2/1**	**11/5**	**11/4**	**14/5**

Table 4d Hexany 4

	Original	Trans. 1	Trans. 2	Trans. 3	Trans. 4	Trans. 5
1:	**33/32**	**35/32**	**77/64**	**21/16**	**55/32**	**15/8**
2:	**35/32**	**1225/1056**	**245/192**	**245/176**	**175/96**	**175/88**
3:	**77/64**	**245/192**	**539/384**	**49/32**	**385/192**	**35/16**
4:	**21/16**	**245/176**	**49/32**	**147/88**	**35/16**	**105/44**
5:	**55/32**	**175/96**	**385/192**	**35/16**	**275/96**	**25/8**
6:	**15/8**	**175/88**	**35/16**	**105/44**	**25/8**	**75/22**

Table 4e Hexany 5

1

Transposed Hexany	2	3	4	5	6	7	8	9	10	11
Pitch Hexany	**1**	**4**	**2**	**3**	**5**	**1**	**4**	**2**	**3**	**5**
Transposition Hexany	**3**	**1**	**3**	**5**	**2**	**2**	**5**	**4**	**4**	**1**

Table 5 Cross-transpositions in *Transposed Hexanies No.2-No.11*

1		1	1	1	1					1	1	1	1		
3				3	3	3	3	3	3	3	3				
5								5	5	5	5	5	5	5	5
7	7	7			7	7			7	7			7	7	

Table 6a Gray Code 1

3		3	3	3	3					3	3	3	3		
5				5	5	5	5	5	5	5	5				
1								1	1	1	1	1	1	1	1
11	11	11			11	11			11	11			11	11	

Table 6b Gray Code 2

7		7	7	7	7					7	7	7	7		
1				1	1	1	1	1	1	1	1				
11								11	11	11	11	11	11	11	11
3	3	3			3	3			3	3			3	3	

Table 6c Gray Code 3

5		5	5	5	5					5	5	5	5		
11				11	11	11	11	11	11	11	11				
7								7	7	7	7	7	7	7	7
1	1	1			1	1			1	1			1	1	

Table 6d Gray Code 4

11		11	11	11	11					11	11	11	11		
7				7	7	7	7	7	7	7	7				
3								3	3	3	3	3	3	3	3
5	5	5			5	5			5	5			5	5	

Table 6e Gray Code 5

Figure 1 Wilson's geometric represenation of the hexany

Example 2 Hexany 5 in standard notation